Title: Observing Sunspots to Measure Solar Rotation

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Introduction: Sun is a star made of He and Hydrogen gasses in Plasma state. Like other stars, the sun also rotates on its axis. Since the sun's surface isn't solid, it has different rotations depending on its regions. It is called differential rotation. Sun rotates fastest near the equator (**about 24 days**) and slowest near the poles (**around 35 days**). In 1612, Galileo first discovered that sunspots accelerate and widen as it moves closer to the center from east and slow down as it moves to the west. He thought sunspot's movement, change in size and shape is caused by its differential rotation [1]. Later in 1859, Richard Christopher Carrington observed that sun doesn't move as a rigid body. In 1863, he provided a quantitative description of differential rotation based on his observations of sunspots [2].

Because sun is made of gaseous plasma, it has different angular velocity in different latitudes [3]. Since $\mathbf{v} \propto \mathbf{r}\boldsymbol{\omega}$, it moves fastest where the rotational radius is greatest (*equator*).

Like earth, sun also has its magnetic field. Because of the differential rotation, Sun's magnetic field lines get twisted. These tangled magnetic field lines come through the surface as sunspots [4]. Sunspots occur on the sun's surface in pairs for this magnetic activity. Sun's temperature is around 10,000° F. Sunspots occur in the sun's photosphere; it has a dark central region known as umbra and its surrounding is called penumbra. Umbra is about 6300° F [5]. Since Umbra is much cooler than the surroundings in photosphere, sunspots appear to be dark patches at the sun's surface - location of strong magnetic fields [6].

In this experiment we'll observe a few sunspots over time in different distances ((latitudes) from the equator and compare each sunstops position changes in longitudes to measure solar rotation rate.

Equations:

Average y = $\frac{all\ y\ values\ for\ the\ sunspot}{total\ number\ of\ y}$;

Here y: distance of the sunspot from the y-axis (vertical) in km for each sunspot.

$\sin\theta = \frac{average\ y}{R}$;

Here R: radius of the Sun (696,340 km).

θ: angle of rotation along the x axis, (between x and Sun's radius, R)]

x(t) = $R\cos\theta\sin(\omega t + \phi)$;

Here x: is the horizontal distance of the sunspot in km,

θ is the angle of rotation along the x axis, (between x and Sun's radius, R),

ω is the angular velocity, t is time and ϕ is phase.

$T = \frac{2\pi}{\omega}$;

Here T is period and ω is angular velocity.

Methods: 1. Online Tracker, 2. Data (Sun's images) from SDO website, or from the UAlberta Observatory (white light filter telescope/Halpha, solar filter for telescope required for vision safety). 3. Scientific Calculator.

i. Since sunspots are around 6300° F (umbra). visible light range is better for sunspots observation. We used HMI Intensitygram (orange) for data collection and collected data with date and time in high resolution images and videos to save data.

ii. By using Online Tracker we first established a Heliocentric Cartesian coordinate system (x,y) keeping the center of the sun at (0,0) for the collected data (video). From the center of the Sun we measured the boundary distance and using a calibrated scale we set the radius, R = **696,340 km** (known value). Based on this, we measured all the data (x,y) in km. Here x is the horizontal distance and y is the vertical distance of the sunspot.

iii. We calculated the Average y and θ for each sunspots.

vi. For the equation x(t) = $R cos\theta sin(\omega t + \phi)$; we plotted the x, y data for time (t) and looked for some ω and ϕ value for the best fit line (with uncertainties) for each sunspot.

vii. Using the ω (best-fitted) value for each sunspot, we calculated T for all the sunspots.

Fig 1.1: Comparison of Observed and Theoretical x(t) for sunspot 1

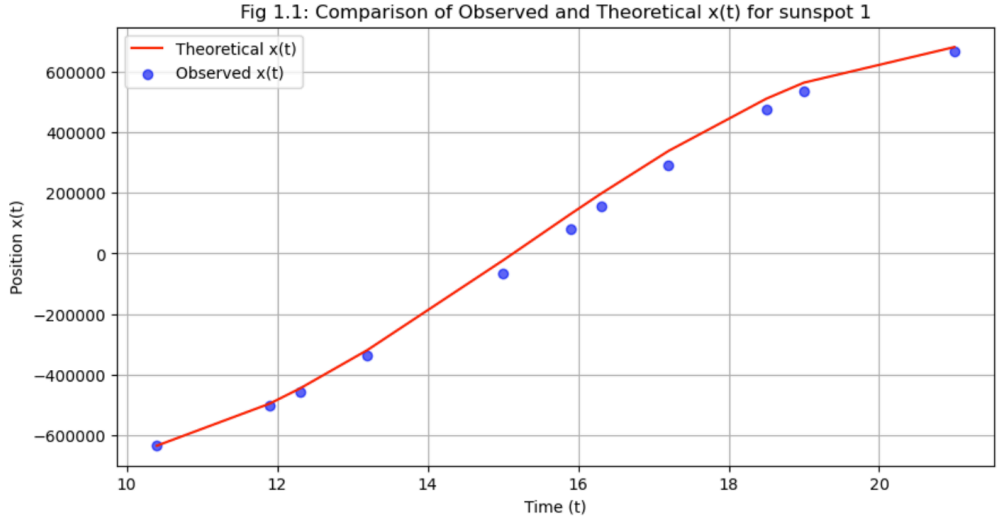


Fig 1.2: Comparison of Observed and Theoretical x(t) for sunspot 2

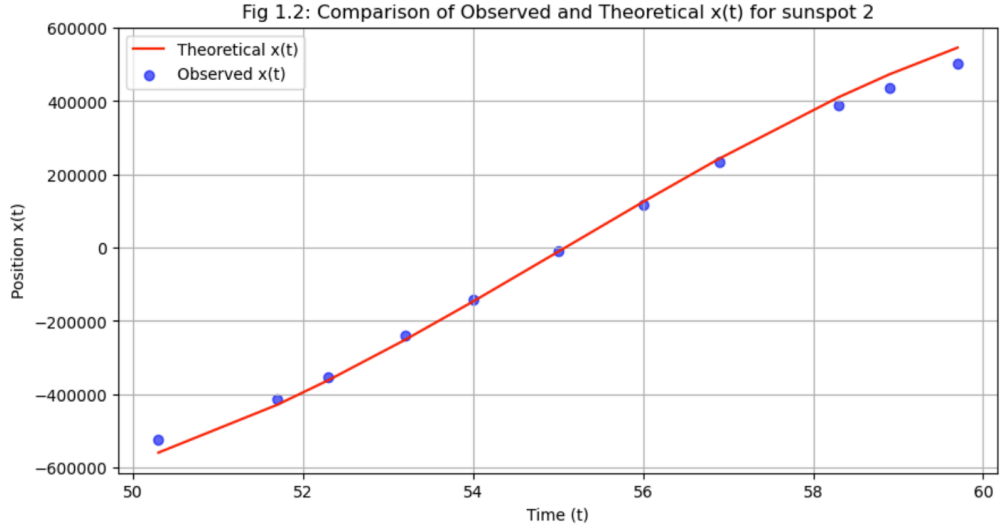
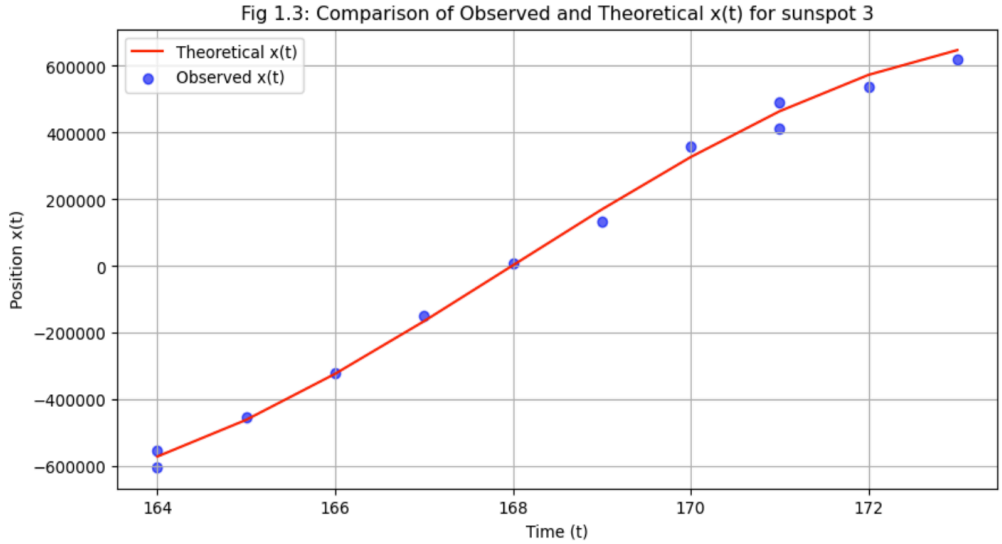
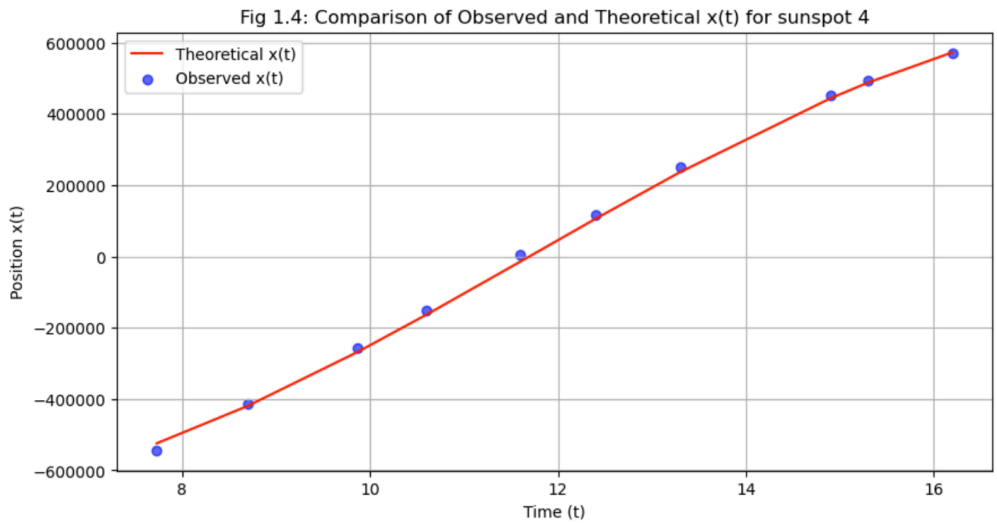


Fig 1.3: Comparison of Observed and Theoretical x(t) for sunspot 3





<u>Fig.1.1, Fig 1.2, Fig 1.3, Fig 1.4</u>: I plotted the x(t) value along the y axis and time (t) along the x axis for each sunspot for the equation x(t) = $R\cos\theta\sin(\omega t + \phi)$; Here **x**: The distance measured along the x-axis (horizontal distance) in km. R: Sun's radius 696,340 km, θ from sample calculation using sinθ = $\frac{average\ y}{R}$, and checked for different values of ω : angular velocity and ϕ = phase for the best fit line.

Sunspot 1: θ = -10.60° , ω =0.25 ± 0.01 rad/s, ϕ = 2.5 ± 0.01 radians
Sunspot 2: θ = 38.03° , ω = 0.2 ± 0.01 rad/s , ϕ = 1.55 ± 0.01 radians
Sunspot 3: θ = 5.68° , ω = 0.247 ± 0.01 rad/s , ϕ = 2.49 ± 0.01 radians
Sunspot 4: θ = 27° , ω = 0.22 ± 0.01 rad/s , ϕ = 3.71 ± 0.01 radians

Sample Calculation:

Sunspot 1: Average y = -1.28E+05

$\sin\theta = \frac{average\ y}{R} = \frac{-1.28E+05}{696{,}340}$ = -1.84E-01, hence θ= $\sin^{-1}$(-1.84E-01) = -10.60°

x(t) = $R\cos\theta\sin(\omega t + \phi)$; x(t) = $696340\cos(-10.60°)\sin(\omega(t) + \phi)$;

I plotted the data in python and adjusted the value of ω and ϕ to check for the best fit line.
ω = 0.25 ± 0.01 rad/s and ϕ = 2.5 ± 0.01 radians is the value for the best fit line.

$T = \frac{2\pi}{\omega} = \frac{2\pi}{0.25 \pm 0.01} = 25.13 \pm 1.007$ days ≈ 25 ± 1 days for sunspots close to the equator.

Sunspot 2: Average y = 4.29E+05

$\sin\theta = \frac{average\ y}{R} = \frac{4.29E+05}{696{,}340}$ = 6.16E-1, hence θ= $\sin^{-1}$(6.16E-1) = 38.03°

x(t) = $R\cos\theta\sin(\omega t + \phi)$; x(t) = $696340\cos(38.03°)\sin(\omega(t) + \phi)$;

I plotted the data in python and adjusted the value of ω and ϕ to check for the best fit line.
ω = 0.20 ± 0.01 rad/s and ϕ = 1.55 ± 0.01 radians is the value for the best fit line.

$T = \frac{2\pi}{\omega} = \frac{2\pi}{0.20 \pm 0.01} = 31.42 \pm 1.57 \approx 31 \pm 2$ days for sunspots 2 is near the pole.

Sunspot 3: Average y = 6.89E+04

$\sin\theta = \frac{average\ y}{R} = \frac{6.89E+04}{696{,}340}$ = 9.89E-2, hence θ= $\sin^{-1}$(9.89E-2) = 5.68°

x(t) = $R\cos\theta\sin(\omega t + \phi)$; x(t) = $696340\cos(5.68°)\sin(\omega(t) + \phi)$;

I plotted the data in python and adjusted the value of ω and ϕ to check for the best fit line.

$\omega = 0.247 \pm 0.01$ rad/s and $\phi = 2.49 \pm 0.01$ radians is the value for the best fit line.

$T = \frac{2\pi}{\omega} = \frac{2\pi}{0.247 \pm 0.01} = 25.44 \pm 0.1$ days ≈ 25 days for sunspots near the equator.

Sunspot 4: Average y = 3.16E+05

$sin\theta = \frac{average\ y}{R} = \frac{3.16E+05}{696{,}340}$ = 4.54E-1, hence $\theta = sin^{-1}$ (4.54E-1) = 27°

x(t) = $Rcos\theta sin(\omega t + \phi)$; -6.33E+05 = $696340cos(38.03°)sin(\omega(t) + \phi)$;

I plotted the data in python and adjusted the value of ω and ϕ to check for the best fit line.
$\omega = 0.22 \pm 0.01$ rad/s and $\phi = 3.71 \pm 0.01$ radians is the value for the best fit line.

$T = \frac{2\pi}{\omega} = \frac{2\pi}{0.22 \pm 0.01} = 28.56 \pm 1.30$ days ≈ 29 ± 1 days for sunspots 4 near the pole.

Result and Discussion: For the sunspot 1 and sunspot 3 near the equator we got period of **25 ± 1** days and **25 days** respectively, which is close to the theoretical period of sunspots near the equator (about **24 days**). Also for sunspot 2 and sunspot 4 we got period of **31 ± 2** days and **29±1** days respectively, which is also close to the theoretical period of sunspots near the poles (about **35 days**).

We used Online Tracker and collected data in high resolution (images/video) to minimize the error in tracking the sunspot's position. Also, we kept backup images (data) to work. We repeated the process a few times for each sunspot and took the average value for calculations to reduce error and biases. We measured uncertainties to minimize the error in calculations. We could have considered the size of the sunspot. Since sunspots' size and shape change over time and it's difficult to keep track of them, we didn't calculate it.

Our expected results are:

1. Theoretically the Sun rotates fastest near the equator (about 24 days) and slowest near the poles (around 35 days), which is true based on our data since Sunspot 1 and 3 take less time (**25 ± 1** days and **25 days)** than sunspot 2 and 4 (**31 ± 2** days and **29±1** days).

2. Also the sun has different angular velocity in different latitudes and it moves fastest where the rotational radius is greatest (*equator*). For sunspots near the north pole our angular velocities are **0.2 ± 0.01** rad/s and **0.22 ± 0.01** rad/s whereas for the sunspot near the equator we got **0.25 ± 0.01** rad/s and **0.247 ± 0.01** rad/s. Hence for our data angular velocity near the equator is greater than angular velocity near the poles.
Hence our results agree with the expected results.

Conclusion: We used an online tracker to set up our coordinate system and scale for the measurement. Based on that we collected data from a video of rotating Sun (SDO website) for four sunspots (2 near the equator, 2 near the north pole). We collected sunspots distance from horizontal and vertical axis with time and using this we plotted the data and looked for the best-fit line for different angular velocities and phases and calculated the period of the sunspots. Periods of the sunspots (which are close to the equator) **25 ± 1** days and **25 days** and (which are close to the poles) **31 ± 2** days and **29±1** days.

Appendix:

Table 1: We tracked the x and y for the sunspots on (t) time each day from
Sunspot 1: 14-01-2023 to 23-01-2023 ; Sunspot 2: 01-03-2023 to 11-03-2023
Sunspot 3: 15–07-2023 to 25-07-2023; Sunspot 4: 11-01-2023 to 20-01-2023
Here **x**: The distance measurement along the x-axis (horizontal distance) in km. **y**: The distance measurement along the y-axis (vertical distance) in km.

Sunspot 1 (close to the equator)			**Sunspot 2** (near the north pole)		
t (s)	x (km)	y (km)	t (s)	x (km)	y (km)
1.04E+01	-6.33E+05	-1.47E+05	5.03E+01	-5.23E+05	4.04E+05
1.19E+01	-5.01E+05	-1.35E+05	5.17E+01	-4.13E+05	4.20E+05
1.23E+01	-4.56E+05	-1.28E+05	5.23E+01	-3.54E+05	4.29E+05
1.32E+01	-3.36E+05	-1.23E+05	5.32E+01	-2.41E+05	4.38E+05
1.50E+01	-6.69E+04	-1.17E+05	5.40E+01	-1.41E+05	4.42E+05
1.59E+01	7.90E+04	-1.17E+05	5.50E+01	-8.87E+03	4.46E+05
1.63E+01	1.56E+05	-1.17E+05	5.60E+01	1.17E+05	4.46E+05
1.72E+01	2.92E+05	-1.22E+05	5.69E+01	2.34E+05	4.41E+05
1.85E+01	4.76E+05	-1.36E+05	5.83E+01	3.90E+05	4.24E+05
1.90E+01	5.36E+05	-1.42E+05	5.89E+01	4.37E+05	4.18E+05
2.10E+01	6.65E+05	-1.70E+05	5.97E+01	5.02E+05	4.07E+05

Sunspot 3 (close to the equator)			**Sunspot 4** (near the north pole)		
t (s)	x (km)	y (km)	t (s)	x (km)	y (km)
1.64E+02	-6.03E+05	9.14E+04	7.73E+00	-5.45E+05	2.97E+05
1.64E+02	-5.52E+05	8.32E+04	8.70E+00	-4.13E+05	3.09E+05
1.65E+02	-4.52E+05	7.37E+04	9.87E+00	-2.57E+05	3.26E+05
1.66E+02	-3.19E+05	5.87E+04	1.06E+01	-1.50E+05	3.30E+05
1.67E+02	-1.48E+05	5.60E+04	1.16E+01	4.26E+03	3.37E+05
1.68E+02	7.51E+03	5.60E+04	1.24E+01	1.16E+05	3.33E+05
1.69E+02	1.32E+05	5.60E+04	1.33E+01	2.52E+05	3.27E+05
1.70E+02	3.57E+05	6.55E+04	1.49E+01	4.52E+05	3.11E+05
1.71E+02	4.11E+05	6.69E+04	1.53E+01	4.93E+05	3.05E+05
1.71E+02	4.89E+05	7.23E+04	1.62E+01	5.71E+05	2.89E+05
1.72E+02	5.36E+05	7.78E+04			

References and Acknowledgements: TA (Aaron Lyons) from the Physics Help Center helped me complete the experiment.

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